

A Review of Fault Diagnostic and Monitoring Schemes of Induction Motors

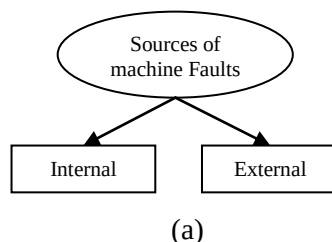
Amandeep Sharma¹, S. Chatterji², Lini Mathew³, Mohammad Junaid Khan⁴

^{*}Electrical Engineering Department, NITTTR, Chandigarh, India

Abstract-Working as the backbone of modern industry, induction machines play a vital role in areas such as manufacturing, transportation, etc. But like any other piece of equipment, they are vulnerable to faults, which if left unmonitored, might lead to catastrophic failure of the machine in the long run. On-line condition monitoring of the induction motors has been widely used in the detection of faults. This paper inspects into the various faults and review of traditional and ingenious techniques for induction motor faults with a recognition of future research areas.

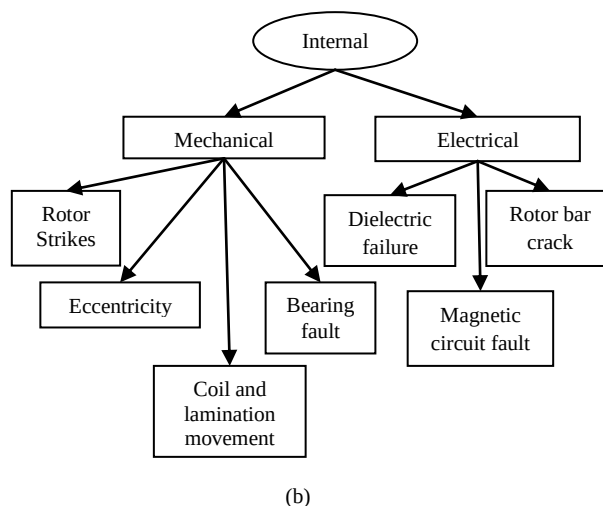
I. INTRODUCTION

Induction motors are considered as foundation for every industry. However they will eventually fail like any other machine because of heavy duty cycles, poor working environment, installation and manufacturing factors, etc. There failure might cause immense amount of loss to owner and clients as well. Consequently, fault diagnosis system is very important for safe operation and preventing rescue. Induction machine failures include mechanical and insulation faults. However, many methods have been proposed for fault detection and diagnosis, but most of the methods require a good deal of expertise to apply them successfully. The induction motor is subjected to major faults and related secondary faults. Various types of faults are categorized into different classifications, as shown in Fig. 1. Mainly, an induction motor may have internal fault or external fault. On the basis of origin, a fault may be classified into mechanical or electrical fault. Depending on the location, fault may be a rotor fault or a stator fault. The faults associated with the moving parts like bearing and cooling are classified as rotor faults, while those associated with the stator are known as stator faults [1]. Specifically, induction motor faults can be broadly categorized into bearing failures, stator faults, rotor faults, air gap eccentricity, mechanical vibrations, etc. Fig. 2 illustrates the relative probability of occurrences of various faults in an induction machine [2].

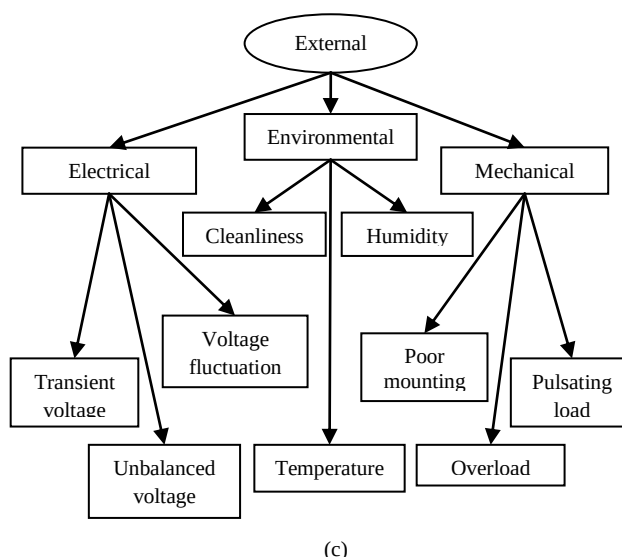


(a) Sources of Machine Faults

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(b) Classification of Internal Faults



(c) Classification of External Faults

Fig. 1 Broad classification of Different Faults in Induction Motors (a) Sources of Machine Faults, (b) Classification of Internal Faults, (c) Classification of External Faults

A fault diagnosis system [3] has been shown in Fig. 3 which consists of a sensor assembly that provides the fault signal to a signal processing unit, which further sends its result to be analyzed by expert systems, where the corresponding fault is ultimately detected.

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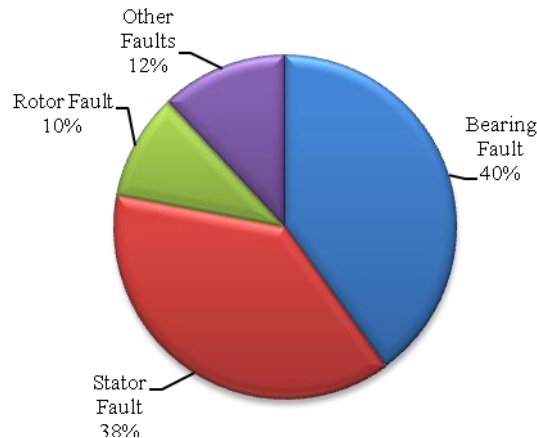


Fig.2. probability of occurrence of various faults [2]

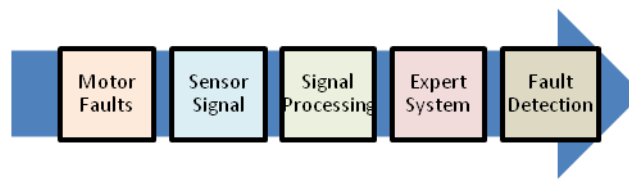


Fig.3. Block diagram representation of On-Line fault monitoring

Condition monitoring and fault diagnostics are practically implemented by the inspection of corresponding anomalies in machine current, voltage and leakage flux. Other methods [4-5] including monitoring the core temperature, bearing vibration level and pyrolyzed products, have been reported to diagnose fault conditions such as insulation defects [6], partial discharge [7] and lubrication oil and bearing degradation [8].

Numerous induction motor fault detection techniques are based on Fast Fourier Transform spectral signature analysis [9-11], vibration analysis [12-13], temperature measurements [14-16], harmonic analysis of speed fluctuations [17-19], state or parameters estimation [20-22], either axial flux or air gap torque analysis [23-25], acoustic noise arrangement [26-27], magnetic field analysis [28-30], fuzzy logic and neural networks [31-33], etc.

II. INDUCTION MOTOR FAULTS

Attentive efforts have been devoted to induction motor fault analysis. Depending on the occurrence of region of fault, induction motor faults are generally put under the following five categories.

A. Bearing Faults

Mainly, a rolling-element bearing is an arrangement of two concentric rings. It is a set of balls or rollers spin in raceways between the inner ring and outer ring. Bearing defects [34] may be divided as “distributed” or “local”. Distributed defects consists misaligned races, waviness, surface roughness and off-size rolling elements. Local defects consists spalls, pits and cracks on the rolling surfaces. These local defects generates a series of impact vibrations at the time when a running roller passes over the surface of a defect whose period and amplitude are calculated by the anomaly’s position, speed and bearing dimension. Mechanical vibrations are produced by the flawed bearings. These vibrations are at the rotational speed of every component. The bearing dimensions and the rotational speed of the machine are used to determine the characteristic frequencies associated to the raceways and the balls or rollers. The condition of the bearing is ascertained by examining these frequencies. This task is accomplished using mechanical vibration analysis techniques.

B. Stator Faults

An induction motor has various stresses like thermal, electrical, mechanical, and environmental [35-37]. Most stator faults can be attributed to such stressful operating conditions. Faults in the stator winding such as inter-turn, inter-coil, open circuit, phase-to-phase and coil-to-ground [38], are some of the more severe and potentially destructive faults. If these faults are left undetected, they may eventually cause cataclysmic failure of the motor.

C. Rotor Faults

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Rotor faults [40] arises due to electrical failures such as a bar defect or bar breakage or mechanical failures such as rotor eccentricity. Bar defect fault occurs due to thermal stresses, hotspots, or fatigue stresses during transient operating conditions such as start-up, especially in large motors. A broken bar changes the torque significantly and became dangerous to the safety and consistent operation of the machines [41]. The second type of rotor fault is caused by air gap eccentricity. This fault is a common effect related to a range of mechanical problems in induction motors such as load unbalance or shaft misalignment. Long-term load unbalance can damage the bearings and the bearing housing and influence air gap symmetry. Shaft misalignment means horizontal, vertical or radial misalignment between a shaft and its coupled load. With shaft misalignment, the rotor will be displaced from its normal position because of a constant radial force.

D. Eccentricity Faults

Unequal air gap between stator and rotor results in eccentricity [42] of induction motor. In general, air-gap eccentricity can be of two types: the static air-gap eccentricity and the dynamic air-gap eccentricity. A mixture of both forms, called mixed eccentricity [43] and the axial non uniformity of air gap, known as inclined eccentricity [44] have also been accounted. The minimal radial air-gap length is fixed in space for static air-gap eccentricity. On the contrary, the center of rotor and the center of rotation do not coincide for dynamic eccentricity. In this case, the position of minimum air gap is not fixed in space but rotates with the rotor. An erroneous positioning of the rotor or stator during the commissioning phase may give rise to static eccentricity. It may also be caused by stator core being non-circular in shape. A cause of dynamic eccentricity can be a bent shaft, bearing wear and movement, or mechanical resonances at critical speeds.

III. FAULT DIAGNOSIS TECHNIQUES

The spectrum of supply current of an induction motor can be analyzed for the diagnosis of faults appearing in it. For a healthy motor, there will be no existence of backward rotating field and only the forward rotating magnetic field rotates at synchronous speed. In the occurrence of any fault, there will be a resultant backward rotating field in the air gap and the spectrum of stator current will change.

In Table 1 various frequency components introduced in supply current spectrum due to faults [45] has been summarized.

Table 1. Frequency components of induction motor fault

Fault	Frequency
Bearing fault	$f_{cbf} = f_s \pm n f_{rc} $ $f_{rc} = \frac{N_b f_s}{2} \left[1 \pm \frac{D_b}{D_p} \cos \theta \right]$
Eccentricity	$f_{ec} = f_s \left[(n R_s \pm O_{re}) \frac{(1-s)}{p} \pm O_{smh} \right]$
Rotor fault	$f_{rbf} = f_s \left[n \frac{(1-s)}{p} \pm s \right]$

where,

f_{cbf} = components generated by bearing faults, f_s = supply frequency, $n = 1, 2, 3 \dots$ [Integral values], f_{rc} = characteristics race frequencies, N_b = number of bearing balls, D_b = ball diameter, D_p = bearing pitch diameter, θ = contact angle of the ball on the races, f_{rbf} = components generated by broken rotor faults, p = number of pole-pairs, s = per-unit slip, f_{ec} = components associated with eccentricity, O_{re} = rotating eccentricity order, O_{smh} = stator MMF harmonic order, R_s = number of rotor slots.

By monitoring the various frequencies in Table 1, it is possible to diagnose the corresponding motor faults. Nevertheless, it has been observed [46-47] that these frequency components cannot be treated as exact signs of occurrence of motor faults as these frequencies can be detected in the spectrum of even healthy motors due to unavoidable manufacturing symmetries and misalignment etc., Many techniques [48-49] have been proposed and developed over years that provide a good measure of occurrence of faults in induction motors.

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A. Vibration Spectrum Analysis

The forces occurring in the rolling element bearing in electrical machines create high frequency components of vibration. During normal conditions, these high frequency components are mainly because of friction but in case of a defect in bearings shock pulses can also be found due to breaks in lubrication layer between the friction surfaces. This method analyses the vibration spectrum of an induction machine using piezoelectric accelerometer which works on Fast Fourier Transform to extract from a time domain signal the frequency domain representation.

B. Using Park Vector Approach and Complex Wavelets

The impact of stator fault on machine current can be examined through Park vector transformation approach. The locus of instantaneous spatial vector sum of the three phase stator currents forms the basis for Park's vector. Generally, a three phase induction motor does not involve the connection to the neutral. It only involves the connection with the three phase mains. Hence, the mains current has no homo polar current. A two dimensional representation of three phase current of induction motor can be achieved using Park's Transform. As a function of mains phase variable (I_A , I_B , I_C), the Park's vector components (I_d , I_q) are:

$$I_d = \sqrt{\frac{2}{3}} I_A - \sqrt{\frac{1}{6}} I_B - \sqrt{\frac{1}{6}} I_C$$

$$I_q = \sqrt{\frac{1}{2}} I_B - \sqrt{\frac{1}{2}} I_C$$

Under ideal conditions, three-phase currents lead to a Park's vector with the following components:

$$I_d = \frac{\sqrt{6}}{2} I_M \sin \omega_s t$$

$$I_q = \frac{\sqrt{6}}{2} I_M \sin(\omega_s t - \frac{\pi}{2})$$

Where, I_m is the supply phase current maximum value and ω_s is the supply frequency. This maps a circle which has its centre at the origin (0, 0) of the coordinates. This is a very simple reference figure that allows the detection of faulty conditions by monitoring the deviations of the acquired patterns. This locus is distorted by stator winding faults and thus provides easy fault diagnosis. The healthy pattern differs slightly from the expected circular one, because supply voltage is generally not exactly sinusoidal.

Besides making analysis and calculations easier, Park's Transform scores over traditional methods in terms of being economical, especially for small and medium-sized motors. Also it can diagnose the faults without requiring access to motor.

C. Motor Current Signature Analysis (MCSA)

Current harmonics in the stator current are analyzed by MCSA [53-59]. These harmonics are caused by new rotating flux components on account of a fault. It needs only one current sensor and is based on signal processing techniques like FFT [60]. The generalized equipment set up for measuring motor current is shown in Fig. 4. Data acquisition is achieved by performing FFT on the stator current. The data obtained after FFT is normalized as a function of the first harmonic amplitude is then analyzed.

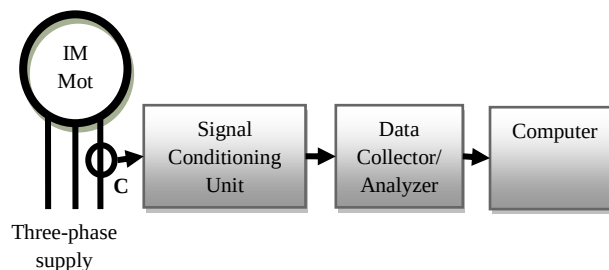


Fig.4 Set-Up for Measuring Motor Current

D. Intelligent Techniques

Several Intelligent techniques like Fuzzy logic systems [61-63], Artificial Neural Networks [64-66], Neuro-Fuzzy Systems [67-69] etc. have been elaborately discussed for induction motor condition monitoring. Usually, any Artificial Intelligence (AI)

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based diagnostic technique has three prime steps- i) Signature extraction ii) Fault detection and iii) Fault severity estimation. Neural Networks for condition monitoring of an induction motor is presented in [70].

Apart from these techniques, some other methods for the incipient fault detection of induction motor have found a mention in other literature [71-75]. Table 2 presents an elaborate comparison of several fault detection techniques and the faults generally detected by them.

Table 2. Comparison of various fault detection techniques

Motor Faults	Detection Techniques
Bearing and Vibration Faults	Vibration Testing and Analysis
Bearing, Rotor, Stator and Vibration Faults	MCSA
Bearing and Stator Faults	Park's Transform
Bearing, Rotor, Stator and Vibration Faults	Wavelet Analysis
Bearing and Rotor Faults	Artificial Neural Networks

IV. CONCLUSION

This paper attempts to summarize recent developments in induction motor fault diagnostics and prognostics. It attempted to review fundamentals, main results, and practical applications of the MCSA used for induction motors faults detection. Various techniques, models and algorithms have been analyzed and the suitability of a particular technique for a specific fault diagnosis has been focused upon in this paper.

Expert systems can be employed for fault diagnosis using rules obtained from the connection weight of a supervised neural network and rules extracted from heuristic knowledge. This combination of Artificial Neural Networks and expert knowledge may enhance the monitoring system for diagnosis.

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